

Image Enhancement using Various Histogram Equalization Techniques

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Abstract—Image Enhancement is a method that converts a poor quality image into a image of superior quality. This paper basically intends to enhance the quality of a image using the Histogram Equalization technique on MATLAB (R2014a) software and the outcome of the image parameters before and after applying Histogram Equalization are discussed. The statistical parameters of the image histogram such as mean and median can be used analyse and show an image of improvised image quality. The method is used to improve the image quality of an X-Ray image of a human bone. This paper also discusses the modern and enhanced techniques of Histogram Equalization and their merits over the traditional Histogram Equalization technique.

Index Terms—Image Processing, Image Histogram, Histogram Equalization.

I. INTRODUCTION

Image Enhancement is a means by which an image can be manipulated to make it more suitable and application specific. Thus the Image must be Enhancement in such a way that it is problem oriented as described in [1]. Like the process of enhancing an Image used for manipulating an image for an Medical application may not be same as that used for Image captured by Satellite. Whatever may be the method used for or the application of Image Enhancement, it is one of the most vital fields in Digital Image Processing as per [2].

Digital Image Processing is one of the most dynamic domain with applications spread out in the field of space exploration, medicine, automated industry inspection authentication and many more areas. The steps included in Digital Image Processing include statistical analysis, manipulation, storing and displaying of graphical image resources. The main objective of Digital Image Processing is to reduce the undesirable noise from the image source. [2] Image Processing is the primary step for the statistical analysis and identification of a particle in an image which is further used for testing in the fields of Digital Photography, Tests and Measurements in the Bio-medical Industry, Microscopic Image Enhancement and many more Digital Image Processing Applications. [3]

Histogram Equalization includes enhancing the quality of the image so that it can be better perceived and interpreted by human beings and other automated image processing systems. This helps in making the image task specific and more suited for the required application. It includes altering the image attributes and parameters as per the demand of the application. After modification of the image, it should be satisfy observer-specific factors such as human visual system and observer's experience. Image Enhancement is used for Noise Reduction in the Image, Enhancement of Dark Images and Improving the contrast of the image [4]. The ultimate goal of Image Enhancement is to convert the original image into an enhanced quality image to make it application specific. [5]

Image Enhancement is a vital section on the field of Digital Image Processing and is very useful in the Medical field, Law Enforcement, Photography, Geographical Information Services (GIS) and various Industrial Applications and hence study and analysis of image quality improved due to Image Enhancement is an essential field in Image Processing . [2]

II. METHODOLOGY

The process with which the image was equalized and then normalized is described here. The Image Enhancement is taking place using the Histogram Equalization technique.

The functioning of the entire process is described in the flowchart in Fig. 2. An X-ray image is taken as the input. The histogram of the image is then plotted. It is essentially the plot of intensity levels of pixels versus the number of pixels. Histogram function will have two inputs. One input is the image itself and the other is the number of discrete instants. Even if the number of instants is not specified in the input arguments, the process will run assuming a default value. The mathematical operations are performed on the image matrix. The function will then convert the matrix into a gray-scale image, assuming black to be the value '0' and white to be the value '1'. The intensity values will then accordingly be adjusted. White and Black will be the maximum and the minimum values respectively. The intensity plot will be broken down for further modifications. The gray-scale image

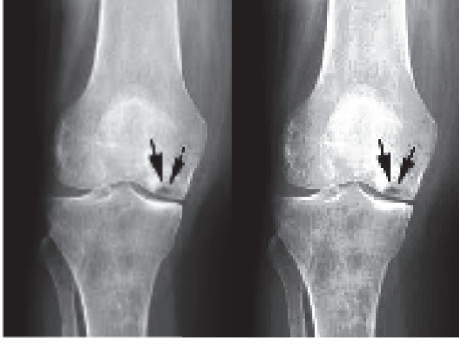


Fig. 1. Comparison of Initial and Final Images after Histogram Equalization Technique

is then converted into an indexed image where the indexed values are specified in the input arguments of the process that are predetermined by the previous task [6]. The indexed image comprises of a array and a colour-map. An indexed image uses direct mapping of pixel values in the array to colour-map. The colour-map is also decided by the previous process. The image is then normalized by conversion into the pdf(probability density function) of pixels. The equalization of pixels is then executed by changing the pdf to cdf(cumulative density function). The attained cdf value converted to pixels by multiplying it by maximum value of range. The inverse of cumulative density function is then calculated and passed through a reshape function with the matrix limits specified and the object is created as in [7]. The output image is then obtained. The histogram is much more spaced out. The intensity levels and the spatial density is improved. The image transitions from an unclear one to a clear one. The image as described in Fig. 1 is brighter due to higher intensity levels.

III. LITERATURE REVIEW

The traditional methods of the Histogram Equalization technique possess certain limitations and fail in certain circumstances. Therefore there is a need to turn to various other modified methods for the Histogram Equalization technique. The modern versions of Histogram Equalization techniques are given below:

A. Adaptive Histogram Equalization Method

The Adaptive Histogram Equalization technique is a type of Histogram Equalization where each image is broken down into tiles. The contrast of the image in each tile is converted into intensity values. If the median value of the image is found in the tile and the gray scale of the window is black, the pixels of the tiles surrounding the 50% gray scale tile are generated first. After this the cumulative gray scale is taken by the sum

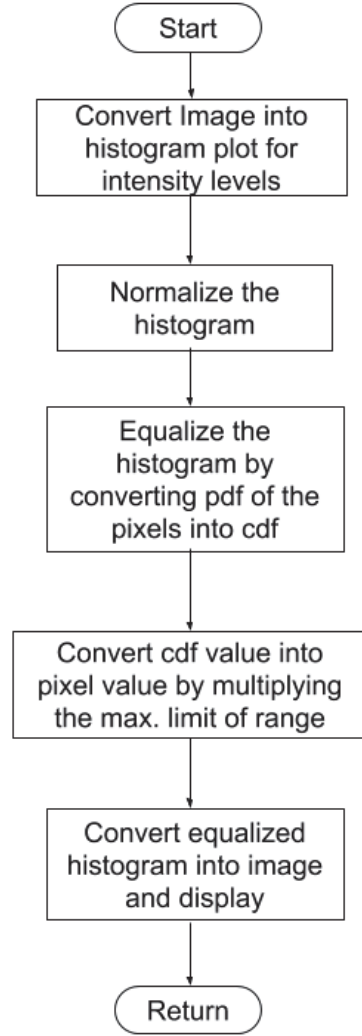


Fig. 2. Flow of steps in the Histogram Equalization Technique

of all gray scale values in the tile and normal procedure of the Histogram Equalization technique is followed further. [8]

As the technique, adapts as per the distribution of gray scale across the image it is named as the Adaptive Histogram Equalization technique (AHE) and is much better than the traditional Histogram Equalization technique.

B. Dual sub-image Histogram Equalization Method

This method of Histogram Equalization divides an image into two sub-images having equal area in the gray scale probability distribution. After the image is divided into two different sub-images, they are equalized individually. After this process is completed, the output is obtained by combination of the two sub-images. [8]

This method along with enhancing the image, brings the gray scale to an average value from extreme values of gray scale. It can be directly used for enhancing a video.

C. Dynamic Histogram Equalization for image contrast Enhancement

The above method includes dividing the histogram of the input image into various image histograms based on the gray scale values. These sub-divided histograms undergo Histogram Equalization independently. This leads to image enhancement in dynamic ranges of gray scale values.

The advantage of this technique is that it avoids the compression of the low contrast grey levels which eventually leads to washing out and blanking of some part of the image. [8]

D. Contrast Limited Adaptive Histogram Equalization Method

In this method of Histogram Equalization there is an algorithm with certain steps that need to be followed.

Firstly the input image is taken which is needed to be enhanced. From this image the number of regions in both the vertical and horizontal direction and number of the bins in a dynamic range in a histogram is found out. The histogram is normalized in the range of 0 to 1 and the clip limit for the image is found out. Further, the image is padded.

The Histogram is made for each of the image in the region with the help of the bin count and it is clipped with a clip limit and converted into a transformation function known as a mapping. A cluster of four of these mappings are formed with them overlapping on each other. Next a single pixel of the image is selected and the four mappings are applied on it. The output of these mappings are interpolated to give the resulting image. [8]

This method helps to bring the high contrast of the image under a certain limit by enhancing the low contrast regions. It is basically a special case of Adaptive Histogram Equalization with a feature to limit the contrast included in it.

These techniques vary their algorithm based on the contrast, the image histogram and the Range of the Gray scale of the input image hence can be used for various types of input images and these at the same time varies different image parameters for different types of image according to the requirement hence are better than the traditional method of Histogram Equalization.

E. Minimum Mean Brightness Error Bi-Histogram Equalization

This technique involves finding the Absolute Mean Brightness Error for different threshold levels of the gray scale. Further, the threshold level with the minimum mean is selected. The histogram is divided into two parts about the threshold value and Histogram Equalization is performed on both the divided histograms.

The other modern Histogram Equalization Techniques give improper results under an unfavourable condition making it a difficult proposition to apply in Consumer Electronics, Medical Applications and other applications where flawless results are expected all the time. The Minimum Mean Brightness Error Bi-Histogram Equalization has no such condition called unfavourable and works without undesirable results throughout all conditions. The only limitation of this technique is that lots

of computation time is necessary to find the Absolute Mean Brightness Error of the various threshold level so that the one with minimum mean is found out.

IV. STATISTICAL ANALYSIS

The statistical analysis of the figure histograms was performed after the execution of the Histogram Equalization process on MATLAB. The parameters that we taken into consideration were the mean, median and the mode which form the basics of statistics. The study was comparative in nature, that is, these parameters were studied after the normalization and the equalization process was completed as in [6] and [7]. The ranges of the mean, median and mode were analyzed. In the figure where the histogram graph of the initial figure has been displayed, the values are being generated at every instant and the ranges are lower spectrum. After equalization, the image was enhanced and the spots where the arrows were indicated as in Fig. 1 tend to be brighter, this makes the scrutiny of X-rays easier for diagnosis purpose. In Fig. 4 we observe that the graph is more spaced out and it is visibly evident that the graph of histogram is like a coalition of discrete values. Histogram is the graph of intensity versus the number of pixels so the discrete values are much higher in the Fig. 4 as compared to the initial plot. Due to Image Enhancement the detection of fractures and faults in the bones will be easier compared to the initial image. The values of the mean in both cases were 103.03 and 129.271 respectively. Taking a look at the values of median for both the histogram figures rendered the results 109 and 125. While distinguishing between the two diagrams in Fig. 3 and Fig. 4, it is noticed that at many distinct discrete points in the equalized figure, the value of the intensity is zilch whereas most of the values in the non-equalized region stay in the bottom part of the spectrum. Hence, when observing the values of mode, the mode value for the non-equalized figure was 29 whereas the value for the equalized figure was zero.

Thus, we check the results of the image before and after applying the Histogram Equalization technique to it. It is observed that the initial image has a low contrast and the histogram of the image does not uniformly spread over the whole range of the gray scale as seen in Fig. 3. The mean, median and mode values of the non-equalized histogram are 103.3, 109 and 29 respectively.

After Histogram Equalization, the histogram of the image spreads over the entire range uniformly and the number of pixels at each gray scale value in the image increases resulting in a high contrast image. Thus the image gets brightened as seen in Fig. 4. Similarly, the mean, median and mode values of histogram equalized image histogram are 129.271, 125 and zero respectively.

However, this traditional method of Histogram Equalization is static in nature without taking into consideration the type of image taken as an input and increases the contrast taking nothing into consideration. For enhancing the image based on the type of the input image, modern Histogram Equalization techniques such as Adaptive Histogram Equalization

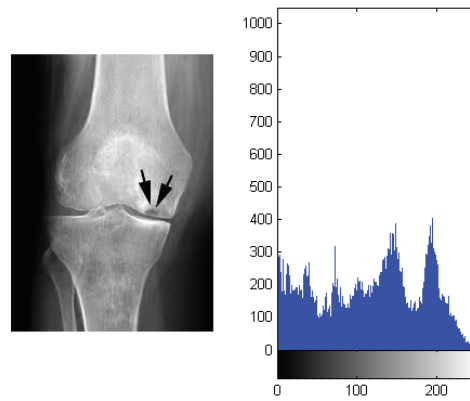


Fig. 3. Histogram of the Image before Histogram Equalization

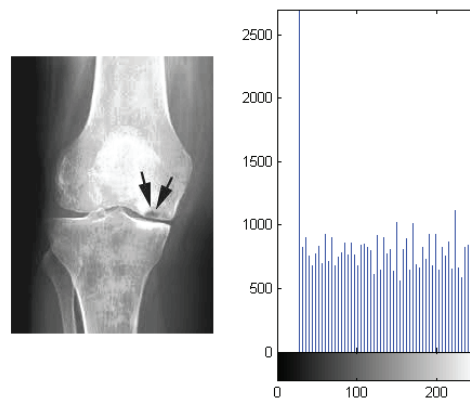


Fig. 4. Histogram of the Image after Histogram Equalization

(AHE), Dualistic sub-image Histogram Equalization (DSIHE), Dynamic Histogram Equalization (DHE) and Contrast Limited Adaptive Histogram Equalization (CLAHE) techniques need to be used. Still these techniques cannot be used in Consumer Electronics and Medical Applications as they give undesirable results under certain circumstances where best results are expected all the time. The Minimum Mean Brightness Error Bi-Histogram Equalization (MMBEBHE) is used for such applications.

V. CONCLUSION

This paper focused on studying Image Enhancement performed using Histogram Equalization Technique. It was performed on an input image using MATLAB (R2014a) software.

The comparisons of the images show that the image after Histogram Equalization is of somewhat higher contrast as compared to the initial image which has a lower contrast. The histogram shows that in the initial image the number of the pixels on the gray scale are close together without spreading over the entire range of the gray scale. After Histogram Equalization it is clearly seen that the histogram

gets distributed entirely over the spectrum of the gray scale, with the number of pixels on each value of the spectrum increasing resulting in the increase in the brightness and the contrast. Comparing the mean values of the histograms of the initial and final image, it is inferred that the mean value of the final histogram is higher than that of the initial histogram. Similar effect is seen on the values of the mode. This shows that the histogram gets spread out over a larger spectrum of gray scale and the image gets brighter. However, when the mode values are compared it is inferred that the value of mode gets reduced from a higher value indicating an increase in contrast of the image.

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